ASTANA IT UNIVERSITY / ICT FINAL PROJECT

DIGITAL QUEUE FOR THE DEAN'S OFFICE

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THE PROBLEM

Students do not know how long the queue will take.

5 / 11

had left the office without being served

91%

would use an electronic queue

Stage 1 survey: n = 11 students

OUR SOLUTION

ONE TELEGRAM CHAT. TWO SIDES.

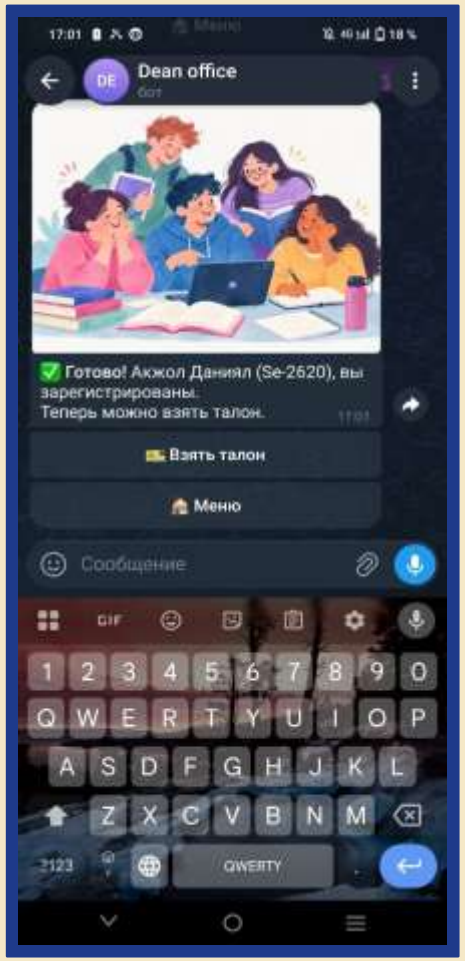
STUDENTS

Take a ticket
See the queue
Get notified

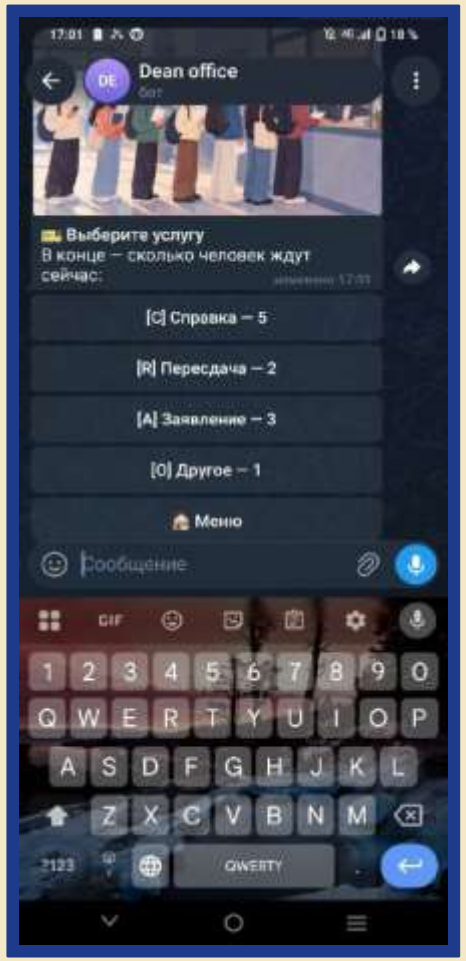
STAFF

Call the next student
Finish or skip tickets
Check statistics

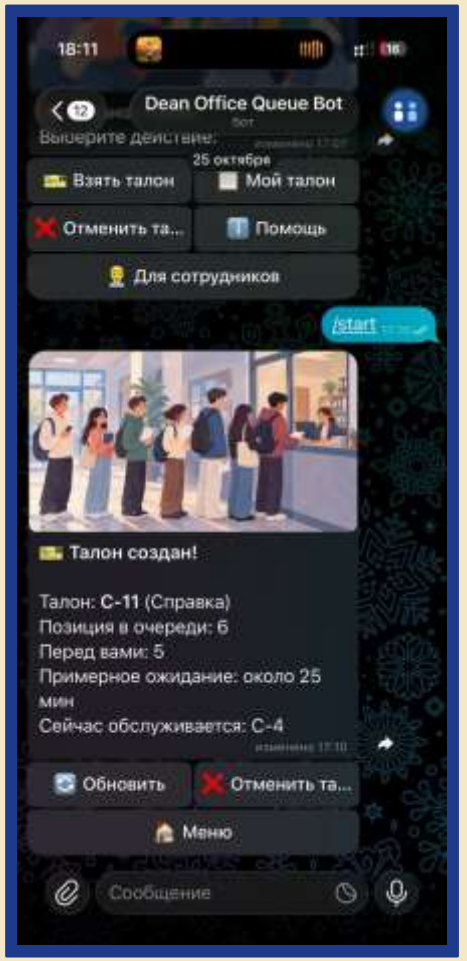
STUDENT JOURNEY



01 REGISTER



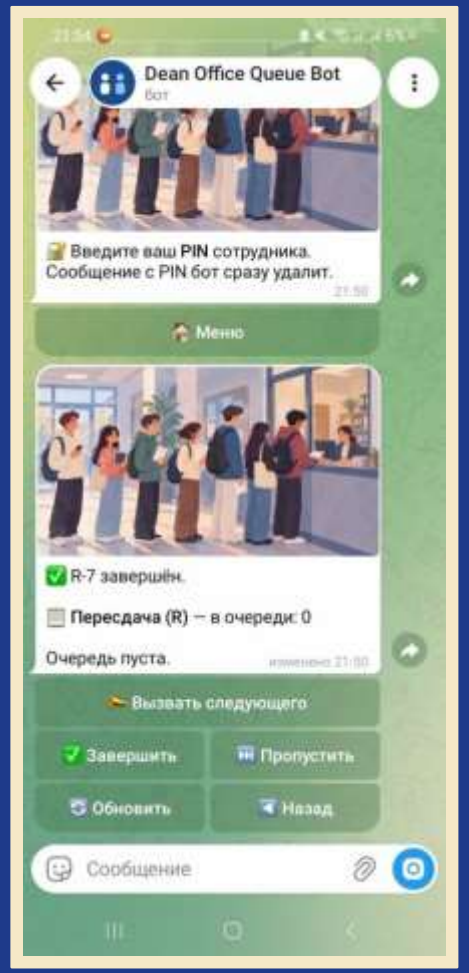
02 SELECT



03 GET A TICKET

STAFF SIDE





PIN LOGIN

Only approved staff can control tickets.

CALL / SKIP / COMPLETE

Stage 2 screenshots: PIN and staff panel

LIVE DEMO

FROM TICKET TO SERVICE

1

STUDENT

Take a ticket

2

OPERATOR

Call next

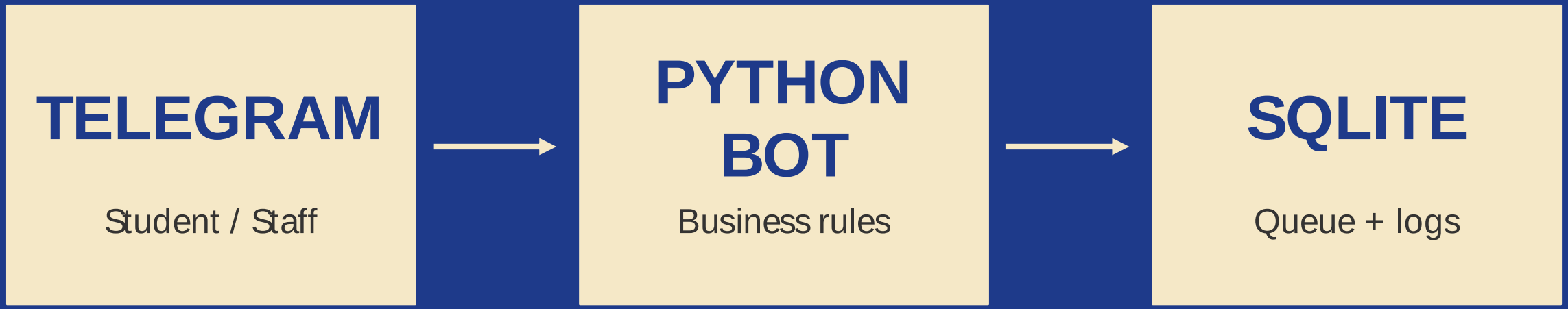
3

ADMIN

View statistics

Backup: Stage 2 screenshots if Telegram is unavailable

HOW IT WORKS



Runs on a team laptop. No public database server.

Simulated corridor sensor → SQLite readings

TEST RESULTS

STAGE 2 MEASUREMENTS

19.3 s

first ticket

3 new users

0.9 s

bot response

5 trials

1.0 s

near-turn alert

5 trials

All 6 student functions passed on two phones.

Final independent re-test is still required.

WHAT WE LEARNED

WORKED WELL

One app for both roles
Simple local database
Fast feedback in tests

STILL LIMITED

Laptop must stay online
No student-system integration
Needs final field testing

NEXT STEPS

01 — RE-TEST WITH 3 USERS

02 — MOVE TO RELIABLE HOSTING

03 — IMPROVE USABILITY

github.com/nuddle227/Telegram-bot

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THANK YOU

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